

# PRATIK DHANAVE

AI Architect

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## PROFESSIONAL SUMMARY

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AI Architect with 7+ years shipping production multi-agent AI systems and cloud-native distributed backends in [Python](#) and [Go](#) across [GCP](#), [AWS](#), and [Azure](#). Hands-on with [LangGraph](#), [Google ADK](#), [Microsoft Agent Framework](#), [MCP](#), [RAG architecture](#), and vector search ([pgvector](#), [Pinecone](#), [AlloyDB AI](#)). Deep Telecom and [Open Banking](#) experience (UAE [ADGM/DIFC](#), Saudi [SAMA](#)); trusted Solution Architect to 15+ enterprise clients on partner-led delivery.

## ACHIEVEMENTS

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- **Architecting production agentic AI for [NESO](#)** — the UK's National Energy System Operator — as AI Architect at [ClearRoute](#) (current role).
- **216 merged PRs (of 270 filed) into Microsoft's official Agent Framework for Go** — runtime correctness across the model-provider, workflow, and agent-runtime layers (highlights [#472](#), [#489](#)); core contributor to [Google's Spanner Migration Tool](#).
- **INR 100 Cr+ saved for [Tata Group](#)** — 57% data-warehouse cost reduction via a [Gemini](#)-powered [BigQuery](#) analyzer, productized as a managed service.
- **30K+ TPS transaction engine and a 1M+ user backend** — led 10 engineers on [Globe Telecom's Kubernetes](#) platform; 47% API latency cut on the [Picnic](#) social network.
- **Speaker, Google Cloud Next 2022** — "[Migrating Monolith Applications into Microservices](#)."
- **Google Summer of Code** — Contributor ([2017–2018](#)), [Mentor \(2019–2026\)](#); guided 10+ students on cloud-native and AI-native projects.
- **105+ published engineering articles** across 15+ series — multi-agent systems, MCP, AI security and governance: [pratikdhanave.com/blog](https://pratikdhanave.com/blog).

## TECHNICAL SKILLS

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**Languages:** [Python](#), [Go](#), [TypeScript](#), [SQL](#)

**Cloud Platforms:** [Amazon Web Services \(AWS\)](#), [Google Cloud Platform \(GCP\)](#), [Microsoft Azure](#)

**AI Engineering:** [Agentic AI](#), [Multi-Agent Systems](#), Agent Workflows, [MLOps](#), [LLMOps](#)

**AI Frameworks:** [LangGraph](#), [LangChain](#), [CrewAI](#), [AutoGen](#), [Google ADK](#), [Microsoft Agent Framework](#)

**AI Protocols & Patterns:** [Model Context Protocol \(MCP\)](#), [A2A](#), [Function Calling](#), [ReAct](#), [Chain-of-Thought](#), [Prompt Engineering](#), [RAG Architecture](#), [GraphRAG](#), Memory Management, [Embeddings](#), [Reranking](#)

**LLM Vendors & Platforms:** [OpenAI](#), [Anthropic Claude](#), [Gemini](#), [Vertex AI Agent Engine](#), [AWS Bedrock Agents](#), [Azure AI Foundry](#), [Hugging Face](#), [Ollama](#), [DataRobot](#) (familiar)

**AI Quality & Ops:** [AI Evaluation \(LLM-as-Judge\)](#), [LangSmith](#), [Guardrails](#), Model Routing

**Vector Databases:** [pgvector](#), [Pinecone](#), [AlloyDB AI](#)

**Databases:** [PostgreSQL](#), [MongoDB](#), [Redis](#), [Cloud Spanner](#), [BigQuery](#), [Azure SQL](#)

**APIs & Protocols:** [REST](#), [gRPC](#), [GraphQL](#), [API Gateway](#), [UPI](#), [Stripe](#)

**Frameworks & Tools:** [FastAPI](#), [Django](#), [Gin](#), [Node.js](#), [Docker](#), [Kubernetes](#), [OpenShift](#) (familiar), [GKE](#), [EKS](#), [Helm](#), [Operators](#), [Terraform](#), [Apache Kafka](#), [Apache Airflow](#), [Prometheus](#), [Grafana](#), [Cloud Run](#), [AWS Lambda](#), [Git](#), [GitHub Actions](#), [Jenkins](#), [Linux](#)

**AI Dev Tools:** [Cursor](#), [Claude Code](#), [GitHub Copilot](#)

**Methodologies & Practices:** [Agile](#), [Scrum](#), [TDD](#), [Spec-Driven Development](#), [Code Review](#), [CI/CD](#), [SRE](#), [Observability](#), [Event-Driven Architecture](#), [Microservices](#), [DDD](#), [CQRS](#), [API-First Design](#)

**Security:** [Cloud IAM](#), [Secret Manager](#), [Cloud KMS](#), [Cloud HSM](#), [Cloud Armor](#), [VPC Service Controls](#), [IAP](#), [Cloud DLP](#), [Binary Authorization](#), [Security Command Center](#), [JWT](#), [OAuth 2.0](#), [OpenID Connect](#), [Entra ID](#), [RLS](#), [Zero-Trust Architecture](#)

## PROFESSIONAL EXPERIENCE

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## AI Architect | [ClearRoute](#)

Jul 2026 – Present

- Spearhead delivery of production agentic AI systems for [NESO](#) (National Energy System Operator) at ClearRoute, a quality engineering and cloud platforms consultancy — operating as a forward-deployed engineer between client and delivery teams.
- Design and implement multi-agent architectures on [Microsoft Agent Framework](#), [LangGraph](#), and [MCP](#) — translating business requirements into scalable designs within real delivery constraints.
- Led the migration of 18 production agents from Google ADK to the Microsoft Agent Framework in 90 days — documented as an [8-part engineering series](#).
- Lead a team of 4 Forward Deployed Engineers; set architecture standards and drive technical decision-making.

## Senior Software Engineer | [Searce Cosourcing Services Pvt Ltd](#) (Google Cloud Premier Partner)

Feb 2022 – Oct 2025

- Led the BigQuery FinOps engagement for [Tata Group](#) — built a [Gemini](#)-powered SQL anti-pattern analyzer (Optimus), refactored queries, and moved to capacity-based slot reservations, delivering **INR 100 Cr+ savings and a 57% data-warehouse cost reduction**; productized as a managed service.
- Architected the [Picnic](#) social platform backend serving **1M+ users** — Go/gRPC microservices behind a GraphQL API gateway, **reducing API response latency by 47%** via [protobuf](#) contracts and service consolidation; led the remote-first team of 4 to 80%+ automated test coverage and 40% lower cloud infrastructure costs.
- Applied AI / AI-Native Engineering team — built Python/[FastAPI](#) services and multi-agent GenAI systems ([Google ADK](#), [MCP](#), [Microsoft Agent Framework](#), [Vertex AI Agent Engine](#)): RAG pipelines, Gemini analyzers, conversational assistants.
- Built and led a team of 4 engineers; mentored 40+ developers across 4 internship cohorts; set architecture standards and drove technical decision-making across product lines.
- Led technical engagement for 15+ enterprise clients as the technical SPOC — reduced onboarding time by 35% and accelerated deployments by 30%; resolved P1 incidents affecting critical client workflows.

### Selected Projects at Searce:

[Litmus](#) — GCP-Powered Industrial IoT Edge Data Platform (Python, Go, [MQTT](#), [OPC-UA](#), [GKE](#), [Pub/Sub](#), [BigQuery](#)) · Senior Agentic AI Engineer · May 2025 – Oct 2025

- GCP-powered backend integrations for Litmus Industrial IoT Edge Data Platform — real-time data ingestion from manufacturing and energy plants (MQTT, OPC-UA), Python streaming pipelines to BigQuery, [Vertex AI](#)-driven anomaly detection, and edge-to-cloud orchestration on GKE + Pub/Sub.

[Bancnet](#) — GCP Agent-Powered Open Banking Portal (Python, Go, [Vertex AI Agent Engine](#), [AlloyDB AI](#), [FastAPI](#), RAG) · Senior Agentic AI Engineer · Aug 2024 – Apr 2025

- GCP-agent-driven API-first Open Banking platform for UAE ([ADGM/DIFC](#)) and Saudi ([SAMA](#)) — [Vertex AI Agent Engine](#) for consent management, data residency, RAG semantic search on AlloyDB AI vector database (37% latency reduction).

[Brownlow](#) — [AFL](#) Voting Platform (Go, Python, GraphQL, gRPC) · Lead Backend Engineer · Mar 2023 – Aug 2023

- Zero-trust voting platform for AFL Brownlow Medal — 100K+ votes, 10K+ concurrent users on Cloud Run; validation, encryption (Cloud KMS), audit logging, abuse prevention via Security Command Center.

[HarbourBridge](#) — Spanner Migration Tool (Go, Python, Cloud Spanner) · Core Contributor · Mar 2022 – Aug 2022

- Core contributor to Google's open-source Spanner migration tool — backend APIs for Intelligent Schema Assistant, CDC pipelines (Datastream/Pub/Sub/Dataflow), 40–60% post-migration query gains via PK design + indexing.

[Globe](#) — Telecom/FinTech Transaction Engine (Go, Python, Kafka, Kubernetes) · Architect & Tech Lead (10 engineers) · Apr 2022 – Sep 2022

- Architected a Kubernetes transaction platform (Go, [Kafka](#), [Pub/Sub](#), [Redis](#)) handling **30K+ TPS** for Telecom/FinTech; **led 10 engineers**; idempotent processing, ledger/reconciliation, dual-layer auth

(JWT + API Keys), PCI-aligned data protection, error-code orchestration with exponential backoff and DLQ fallback.

## Specialist Application Developer | Net Connect Pvt Ltd

Jun 2021 – Feb 2022

### Selected Project at Net Connect:

[azure-service-operator](#) — Kubernetes-Native Cloud Lifecycle for Ericsson & AT&T (Go, Python, Azure, Kubernetes) · Open Source Developer & Architect · Jun 2021 – Feb 2022

- Served as Open Source Developer and Architect for [airshipit](#) and [azure-service-operator](#) in collaboration with [Ericsson](#), [AT&T](#), and Microsoft.
- Integrated OpenTelemetry observability framework into the [airshipit](#) project, reducing manual operations by ~30% and improving incident detection and recovery time.

## Software Developer | CloudCover Pvt Ltd (Google Cloud Premier Partner)

Oct 2019 – May 2020

### Selected Project at CloudCover:

Bloom ([SC Ventures](#)) — Secure Cloud Provisioning for [Standard Chartered](#) ([Terraform](#), AWS, Azure, [SOC 2](#), [ISO 27001](#)) · Cloud Engineer · Oct 2019 – May 2020

- Contributed to Bloom (backed by [SC Ventures](#), Standard Chartered) — built Terraform-based, policy-driven secure cloud provisioning for regulated financial institutions on AWS/Azure with SOC 2 and ISO 27001 controls; developed push-to-deploy self-service capabilities, cutting onboarding time by 20%.
- Designed multi-cloud landing-zone patterns for regulated banking workloads; introduced policy-as-code guardrails ([Open Policy Agent](#)) and centralized audit logging.

## Software Developer | SoftCorner

Feb 2016 – Dec 2017

### Selected Project at SoftCorner:

[Shelter Associates](#) — Open-Source Data Platform for Slum Mapping (Python, [KoBoToolbox](#), [Google Maps](#), [BIRT](#)) · Full-Stack Developer · Feb 2016 – Dec 2017

- Built an open-source data platform for [Shelter Associates](#) — integrated KoBoToolbox for survey data ingestion, built Python data pipelines, automated reporting via BIRT, and delivered spatial data visualization on Google Maps.
- Supported NGO-led slum mapping and rehousing programs across Pune, Sangli, and Kolhapur — delivered geospatial dashboards that informed municipal policy decisions and household-level housing interventions.

## OPEN SOURCE CONTRIBUTIONS

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- [Gocloud](#): Developed a unified API library for managing cloud resources across AWS, GCP, and Azure.
- [CX Programming Language](#): Enhanced CX interpreter for deterministic blockchain transaction execution; implemented [Pratt parser](#) reducing parsing complexity by ~30%.
- [Spanner Migration Tool](#): Redesigned schemas to prevent write hotspots; built CDC pipelines ([Datastream](#), [Pub/Sub](#), [Dataflow](#)) enabling minimal-downtime migrations to [Cloud Spanner](#).
- [Microsoft Agent Framework Go](#): 216 merged PRs (of 270 filed) across the model-provider, workflow, and agent-runtime layers, each with race + regression tests — highlights: tool-approval nil-panic hardening (#472), route-builder handler validation (#489).

## PERSONAL PROJECTS

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[Genie](#) — Azure AI Foundry-Driven Multi-Agent Financial Assistant (Python, Go, [Azure AI Foundry](#), [Microsoft Agent Framework](#), [Postgres](#), [OpenTelemetry](#)) · Author & Lead Engineer · Nov 2025 – Jun 2026

- 15 role-specialized agents (supervisor, analyzer, forecaster, anomaly\_detector, recommender, llm\_auditor, etc.) on Azure AI Foundry + Microsoft Agent Framework; [JWT](#) + [RBAC](#), [AES-256-GCM](#) envelope encryption, end-to-end [OpenTelemetry](#) tracing across agent hops.

[Bodh](#) — Microsoft-Powered [MAI-DxO](#)-Style Medical Diagnostic Agent Panel (Python, Go, [Microsoft Agent Framework](#), [FHIR R4](#), [SD-Bench](#)) · Author & Lead Engineer · Nov 2025 – Jun 2026

- Virtual physician panel with role-specialized agents (intake, supervisor, questioner, test\_planner, cost\_guardian, diagnostician, reasoning\_verifier); FHIR R4 + [HL7 v2](#) aware; cost-aware diagnostic budget enforcement.

## CONSULTING PORTFOLIO

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**Industries:** UAE/Saudi Open Banking · Telecom (Philippines) · FinTech · Energy · Sport · Industrial IoT · Healthcare

**Engagement Types:** Solution architecture · Cloud migration · FinOps · GenAI platform design · Multi-agent system delivery · Open-source upstreaming · Technical SPOC for enterprise accounts

**Client Tiers:** UK critical national infrastructure ([NESO](#)) · Fortune 500 ([Tata Group](#), [Globe Telecom](#), [Dainik Bhaskar](#)) · Tier-1 regulated banks ([Standard Chartered](#) via [SC Ventures](#)) · UAE/Saudi regulated FinTech (Bancnet — ADGM/DIFC/SAMA) · Global telco vendors ([Ericsson](#), [AT&T](#))

**Scope:** 15+ enterprise engagements across 5 countries · Case studies: [pratikdhanave.com/architecture](https://pratikdhanave.com/architecture) · 13 public client & colleague recommendations: [pratikdhanave.com/recommendations](https://pratikdhanave.com/recommendations).

## CERTIFICATIONS

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- [LinkedIn Learning](#): Hands-On AI: Building Agents with the Google Agent Development Toolkit (Google ADK) — Jun 2026.
- [Udemy](#): AWS Certified Generative AI Developer – Professional.
- [Coursera](#): Google Cloud — Generative AI Leader Professional Certificate.
- [Ardan Labs](#): Ultimate Go and Ultimate Services Go.

## EDUCATION

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**Bachelor of Engineering in Information Technology**

2015

[Pune Institute of Computer Technology](#), University of Pune, India

## LANGUAGES

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English (Professional — full business fluency, written and spoken) · Hindi (Native) · Marathi (Native)

## RELOCATION & AVAILABILITY

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- **Open to India (Pune, Bengaluru, Hyderabad, Mumbai, Chennai) and international relocation — Dubai/UAE, UK, Ireland, Netherlands, Germany, Switzerland, Norway, Sweden, Denmark, Finland, Canada, USA, Singapore, New Zealand.** Passport valid through 2028, ready for work-visa sponsorship. Available within 30 days — documents ready for attestation.